Barath B

Android Engineer
barathjack77@gmail.com +91 80159 31576 Trichy, India github.com/3dbarath

Summary

Android engineer who ships complete products alone: a live institutional app serving 1,200 daily users, a Kotlin Multiplatform file transfer tool on Play Store, Windows, Linux and AUR, and the release tooling behind three shipped projects. Spent a summer at Thinkster Math automating worksheet QA with Selenium and LLM grading. B.E. CSE student at GCE Srirangam, expected 2027.

Experience

Thinkster Math Remote, United States
AI Systems & Analytics Intern May 2026 – Jun 2026
Built an AI-powered worksheet automation and analytics system for the Thinkster math platform, used by the product and analytics team to replace manual worksheet QA.

- Selenium + Python framework runs the full worksheet flow unattended: login, worksheet selection, question capture, answer validation and report generation
- LLM analysis over captured question screenshots verifies math correctness, checks answer keys, flags duplicate options and ambiguous questions, and returns ranked fixes
- Prompt engineering on the grading prompt raised response consistency across worksheet types and cut false positives that manual review had to re-check
- Flask reporting app on MongoDB stores every run, screenshot, AI response and issue, with filtering and Excel export for the analytics team
- Thread-safe runner with a shared state lock and graceful stop, so several worksheet batches run in parallel and report live progress instead of blocking

Python, Selenium WebDriver, LLM APIs, prompt engineering, MongoDB, Flask, openpyxl, Git

Projects

GCES Hostel Management System Jan 2026 – Mar 2026
Android + Spring Boot, live on Play Store, 1,200 daily active users.

- Multi-module Clean Architecture across :app, :server and :common, with Hilt scoping ViewModels, repositories and singleton managers
- Predictive meal engine with ShedLock distributed JDBC locks, so midnight state transitions stay atomic across server instances

Kotlin, Jetpack Compose, MVI, Hilt, Retrofit, Coroutines/Flow, Room, Spring Boot 3.5, PostgreSQL, Flyway, JWT, ShedLock, Bucket4j, Firebase, AdMob

ConnectLnx Dec 2025 – Jan 2026
Kotlin Multiplatform, shipped to Play Store, Windows, Linux and AUR. Closed beta Dec 2025 - Jan 2026.

- Custom binary TCP protocol with parallel multi-socket streaming, chunking files across up to 5 concurrent connections for maximum LAN throughput
- Zero-config discovery over mDNS on Android (NsdManager) and Desktop (jmDNS), with QR pairing as the network-restriction fallback

Kotlin Multiplatform, Compose Multiplatform, Ktor, Coroutines/Flow, mDNS, ML Kit, FileKit, multiplatform-settings
github.com/3dbarath/ConnectLnx

Aztrox Release Apr 2026
Internal release CLI built because no existing tool understood the full Aztrox stack.

- One CLI manages build, sign and distribute for 3 live projects across 6 targets: Google Play, GitHub Releases, AUR, Render, HuggingFace Spaces and Firebase
- LIFO atomic rollback registry reverts version bumps, gradle.properties and git tags when a pipeline stage fails, without manual cleanup
- Preflight checks validate dependencies, keystore passwords, env vars and paths before any build; every release is logged to SQLite with generated HTML audit reports

Bash, SQLite, Fastlane, GitHub Actions, jarsigner, gh CLI, ADB

Skills

AI & Automation LLM APIs, Prompt Engineering, Selenium, Python, MongoDB, Flask

Mobile Jetpack Compose, MVVM/MVI, Clean Architecture, Hilt/Dagger, Coroutines & Flow, Room, Retrofit + OkHttp, DataStore, Kotlin Multiplatform, Compose Multiplatform, Firebase

Backend Spring Boot 3.5, Spring Security, PostgreSQL, Flyway, JPA/Hibernate, JWT, ShedLock, Bucket4j, Caffeine, Docker

DevOps & Tooling Git, Fastlane, GitHub Actions, AUR packaging, Bash, SQLite, jarsigner, ADB

Education

Government College of Engineering, Srirangam B.E. Computer Science and Engineering • CGPA 7.6
Trichy, India • Expected 2027